Mohammad (Irmaan) Mirzanejad

PhD Researcher in Artificial Intelligence

Sequential Decision-Making under Uncertainty| Reinforcement Learning | Causal Identification | Reliable & Scalable AI

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Expected PhD graduation: Dec. 2027 **Availability:** Full-time for a research internship in 2027

Research Profile

PhD researcher at CNRS UMR 6072 (**GREYC**) / Université de Caen Normandie. My thesis focuses on **Sequential Decision-Making under Risk**, spanning **tractable** rank-dependent planning, **optimality certification**, risk-sensitive learning under **unknown dynamics**, and extensions toward deep and infinite-horizon settings. Beyond this thesis core, I have explored broader questions in sequential decision-making, including **planning under complete observation loss** and follow-up work on blind planning under persistent model uncertainty. My broader research portfolio spans **reliable reinforcement learning, large-scale structured MDPs, learned-world-model reliability, adaptive evaluation of autotelic and self-expanding agents, causal identification, and efficient LLM inference**, including **five sole-authored** manuscripts submitted to **ICLR 2027** and an AAAI 2027 paper advanced to Phase 2. Across these threads, I emphasize explicit problem formulation, theoretical guarantees, controlled falsification, careful empirical validation, and reproducible research engineering.

Thesis Risk-Sensitive Research Trajectory: tractable model-based surrogate planning → scalable original-MDP optimality certification → unknown-dynamics learning → decision-sufficient distributional RL → *continual risk learning under irreversibility.*

Research Contributions and Ongoing Research

ERQDP: Exact Rank–Quantile Dynamic Programming CNRS GREYC, 2024–present

- Developed a tractable framework for risk-sensitive MDPs whose policies are evaluated by the **full return distribution** under root/resolute CVaR and WOWA, where ordinary Bellman optimality is generally unavailable or intractable.
- To avoid combinatorial trajectory/policy enumeration, formulated a **rank–quantile surrogate** that admits dynamic programming, together with **deterministic distributional evaluation** of the returned policies; resulted a paper in **UAI 2026**.

Optimality Certification for Resolute Risk-Aware Planning AAAI 2027 – Phase 2 review, 2026

- Extended ERQDP from surrogate planning to **deterministic lower/upper optimality certification on the original MDP objective** for resolute rank-dependent CVaR and WOWA trajectory criteria.
- Achieved **0.1% median Original-MDP optimality certificates** across all benchmarks, with a worst observed gap below **1.5%**, while remaining tractable on a model with roughly **470K transitions**.

Certified Risk-Sensitive Planning under Unknown Dynamics ongoing thesis thread, 2026

- Extended ERQDP to the setting where transition dynamics are **unknown and must be learned from finite samples**, so planning is performed on an imperfect empirical MDP rather than an exact model. Built an end-to-end loop— **samples → empirical MDP → ERQDP policy → true-model audit**— together with decision-focused acquisition that prioritizes transition uncertainty according to its downstream influence on the risk-sensitive decision.
- Propagated finite-sample transition uncertainty **through the rank-slice certification machinery**; held-out experiments produce systematically shrinking certificates, roughly **34× tighter** in median than a generic global-transfer bound.

Blind Planning under Missing Observations and Persistent Uncertainty SUM 2026 + follow-up, 2026

- **Governed Blind Commitment Planning:** formulated planning when observations disappear during a commitment window as an auditable constrained optimization problem: enforce explicit harm budgets first, then rank feasible fixed action sequences by mean/tail risk; when none is feasible, return a least-violating plan and diagnose the source of failure.
- **Follow-up work** asks whether a fixed blind sequence can be **structurally desensitized** so persistent transition-model errors oppose and cancel rather than simply being tolerated; formulated exact sequence-level cancellation, established a complexity boundary, and derived a tractable position–action sensitivity decomposition with convex balancing and deterministic rounding.
- Confirmatory and negative-control studies identify when cancellation helps and when low sensitivity is insufficient because nominal sacrifice, common-mode error, or higher-order effects dominate.

Think BIG: Bellman Intervention Geometry for Search-Free Graph Control ICLR 2027 – under review, 2025–2026

- Introduced **Bellman Intervention Geometry (BIG)**, which solves a deliberately small stochastic graph MDP, converts optimal action values into no-op-referenced Bellman lifts, and compiles intervention support, magnitude, and ordering into a frozen scorer that transfers to much larger graphs **without target-state enumeration or online search**.
- Across four stochastic graph-control families, BIG achieves a strong quality–computation regime. On Firefighter, it **strictly Pareto-dominates every measured official PROST return/time operating point**, with a **502–9706×** operation-time separation; on the matched 120-node target, BIG's complete decision path requires about **6400× less measured latency** than SymNet2's forward-only inference. An exact-teacher audit recovers the optimal action in **98.74%** of multi-action states, with no observed violation of the stated 2ϵ regret bound.

Finite-Batch Policy-Gradient Reliability as an Identification Problem ICLR 2027 – under review, 2025–2026

- Formulated finite-batch policy-gradient reliability as an **identification problem**: does the information conventionally retained about a trajectory-gradient estimator determine the probability that a finite batch points in the population-wrong direction? Proved that even bounded REINFORCE problems matching rich conventional descriptors and complete first/second-order policy-gradient functions can have wrong-direction risks separated by **any prescribed factor**.
- Derived a **sharp ambiguity law** even after conditioning on likelihood-level information, constructed genuinely sequential MDPs with distinct directional large-deviation exponents despite matched descriptors, and developed a **non-oracle finite-sample reliability certificate** based on an independent confidence cone and held-out one-sided MGF information. Controlled rare-event studies and an independently confirmed power-grid experiment support the resulting regime-dependent reliability picture.

Prospective Target Preservation for Self-Expanding (Autotelic) Agent Evaluation ICLR 2027 – under review, 2026

- Formalized **candidate-induced target endogeneity**: when an agent or update creates, unlocks, or admits new tasks, those consequences can change the very objective used to decide whether that same candidate is accepted. Proved that any nontrivial same-round target change can reverse some bounded threshold decision **even under exact, noiseless evaluation**.
- Proposed **Prospective Target Preservation**: **freeze → acquire → transport → decide → release**. Changing only the target semantics produced false-promotion rates of **73.4%** and **54.1%** in exact primary and confirmation studies. The mechanism also persisted with naturally trained Procgen policies: an independent replication produced **18.8% false promotions**, including 11 reversals jointly separated from zero under the 95% certificate.

Certified Compression for Causal Partial Identification with Instrumental Variables ICLR 2027 – under review, 2026

- Introduced **target-preserving stochastic representation reduction** for finite instrumental-variable models: outcome support can be compressed while preserving designated causal mean targets model-by-model, treatment-response restrictions, and one-sided identified-set validity. Established when such compression is valid and essentially unique, why nontrivial deterministic compression cannot provide the same guarantee, and how nested refinement monotonically recovers the original finite-support problem.
- The certified reduction yields substantial computational gains without sacrificing the target or one-sided validity: representative multivalued-IV problems obtain **8.0–18.7**× speedups, while compressing an outcome support from $n = 512$ to $M = 32$ gives a median **236× speedup** against the fastest certified exact backend. The framework also extends to finite-sample inference and retains its validity on a public job-training benchmark.

Sparse-Attention Support Selection in LLMs (SwapCert) ICLR 2027 – under review, 2025–2026

- Established a structural limit of rank-then-retain sparse attention: even **exact objective-matched singleton importance** does not generally identify the best fixed-budget support after softmax renormalization. An explicit Top-2 construction incurs **133.2**× the downstream KL of the jointly optimal pair, and an exact exchange analysis proves support-conditioned preference reversals for every $K \geq 2$.
- Exact audits across three LLM families show that singleton Top-K leaves at least **10% recoverable headroom in 164/180** primary prompt–head cases. **SwapCert** provides a mathematically sound same-budget one-swap certificate of remaining headroom and identifies **157/164 (95.7%)** of those exact positive cases, with **zero soundness violations**. This line evolved from earlier Marginal Utility Indices work on dense-reference sparse-computation scoring.

Proof-Carrying Interventional Planning and Decision-Sufficient Causal Learning ongoing research direction, 2025–present

- Developing intervention-aware planning with explicit **positive achievability and negative obstruction certificates**, and formal causal semantics linking operational interventions to changes in underlying mechanisms rather than using "intervention" only descriptively.
- Investigating a decision-sufficiency principle: **learn only the causal structure needed to certify the downstream long-horizon decision**. The learner maintains a causal-model uncertainty set, stops when all remaining models agree on a certified plan/impossibility result, and otherwise chooses interventions to reduce *decision-relevant* ambiguity rather than recovering an entire causal graph.

Structured Regret Amplification in Learned World-Model Planning ongoing research, 2026

- Identified a failure mode in which broader adaptive search amplifies downstream decision regret beyond generic noisy maximization; introduced a matched residual-reassignment null that isolates candidate-specific error structure.
- A frozen prospective study confirmed **excess regret severity beyond the matched null** ($p < 0.001$) across independent pipelines, scorer heads, and planning horizons; failed hypotheses and mitigations were retained as negative evidence.

Decision-Sufficient Distributional RL under Rank-Dependent Risk current thesis frontier, 2026–present

- Investigating whether risk-sensitive decisions require the **full learned return distribution**, or only a small objective-specific set of quantile/rank evaluations sufficient to preserve action choice. Current results show that sparse objective-aware rank queries can closely preserve dense distributional decisions while reducing inference cost; ongoing work develops decision-preservation guarantees, broader risk objectives, and harder benchmarks.

Infinite-Horizon Risk-Sensitive Planning current thesis frontier, 2026–present

- Extending rank-dependent planning and certification beyond finite horizons toward **absorbing stochastic-shortest-path / infinite-horizon settings**, with explicit control of truncation and residual-tail error.

Earlier Research: Multi-Objective RL, Fair Ranking, Applied ML 2020–2023

- Research on Multi-Objective Reinforcement Learning and Multi-Criteria Decision Making; subsequent R&D such as Deep-MORL for tax evasion continual learning, AI-aided tax auditing, and smart meter anomaly detection via ML methods.

Publications and Manuscripts

- **I. Mirzanejad**, N. Bourdache, A.-I. Mouaddib. *Long-Term Sequential Decision Making Under Risk.* **UAI 2026**; PMLR.
- **I. (M.) Mirzanejad**, N. Bourdache, A.-I. Mouaddib. *Planning under Null-Observation Windows.* **SUM (Athens) 2026**; LNAI.
- **M. Mirzanejad**, P. Vamplew, M. Ebrahimi, H. Veisi. *An Online Scalarization Multi-Objective Reinforcement Learning Algorithm: TOPSIS Q-Learning.* **KER, Cambridge University Press**, 37:e7, 2022.

2027 conference manuscripts under review

- **AAAI 2027 – advanced to Phase 2**: *Certified ϵ-Optimal Planning for Resolute Risk-Aware Sequential Decision-Making.*
- **ICLR 2027**: *When Is a Policy-Gradient Batch Reliable? Identification Limits Under Severe Uncertainty.*
- **ICLR 2027**: *Think BIG: Bellman Intervention Geometry for Search-Free Graph Control.*
- **ICLR 2027**: *When Agents Change Their Own Test: Prospective Target Preservation Under Candidate-Induced Task Expansion.*
- **ICLR 2027**: *Preserve Target, Relax Sharpness: Certified Compression for Scalable Instrumental Variable Partial Identification.*
- **ICLR 2027**: *Ranking Is Not Support Selection: Exact Marginals and Certified Headroom for Sparse Attention in LLMs.*

Peer Review

Engineering Applications of Artificial Intelligence (3 reviews, 2025–2026);
Neural Computing and Applications (5 reviews, 2025–2026);
AAAI 2027 (1 sub-review).

Education

Ph.D. in Artificial Intelligence **2024–2027**
Université de Caen Normandie / CNRS UMR 6072 GREYC *France*

- Thesis: **Sequential Decision-Making Under Risk**. Supervisors: Prof. Abdel-Illah Mouaddib and Dr. Nadjet Bourdache.

M.Sc. in Computer Science – Decision Science & Knowledge Engineering **2017–2020**
University of Tehran *Iran*

- Thesis: *Multi-Objective Reinforcement Learning Based on TOPSIS*; thesis grade: **20/20**.
- Coursework: Machine Learning, Statistical & Probabilistic Learning, , Deep Learning, Data Mining, Game Theory, Multi-Criteria Decision Making, Fuzzy Decision Systems.

B.Sc. in Computer Engineering – Software **2006–2011**
Payame Noor University *Iran*

- Final project: fuzzy data storage and retrieval for uncertainty management; grade: **20/20**.

Technical Skills

Programming	**Python, C/C++, C#, Java**; object-oriented design; numerical, scientific, and constraint programming; automation code.
ML / AI	Reinforcement learning, Deep Learning, policy gradients, risk-sensitive and multi-objective RL, structured/factored MDPs, causal identification, optimization under uncertainty, deep learning, world models, sparse attention/LLM inference, computer vision, recommender systems, probabilistic modeling.
Frameworks / Libraries	**PyTorch, JAX, TensorFlow, Keras**, scikit-learn, Gymnasium; NumPy, pandas; experiment/config pipelines and automated testing.
Research Engineering	Parallel and distributed computing; large-scale experiment orchestration; reproducible research pipelines and experiment provenance; probabilistic planning benchmarks (RDDL/PROST); exact enumeration; permutation/bootstrap inference; finite-sample statistical analysis; automated testing and evaluation pipelines; agentic workflow orchestration.
Robotics	ROS, testing mobile robots, drones/UAVs, cameras and sensors, HRI scenarios, multimodal QA.
Web / Data	ASP.NET, Java/Spring, Django, Node.js, ReactJS, PHP/Laravel, JavaScript, HTML/CSS/Bootstrap; REST APIs; MSSQL, MySQL, Oracle, MongoDB.
Systems / Cloud	Linux/Windows servers administration; virtualization, routing/switching, BGP/OSPF, firewalls; Azure and AWS networking; Terraform Enterprise; monitoring, incident response, security hardening.
Cybersecurity	Penetration testing, malware/exploit analysis, network security, cryptography/security protocols, IDS/IPS/VPN/TLS concepts.

Professional and Engineering Experience

PhD Researcher – Artificial Intelligence **2024–present**
CNRS UMR 6072 GREYC / Université de Caen Normandie *Caen, France*

- Design algorithms, proofs, experiments, and scalable reproducible research pipelines for sequential decision-making under uncertainty, particularly risk-sensitive planning, validation and learning with possible extension to continual risk learning.

Data Science Expert **2022–2024**
Iranian National Tax Administration *Tehran, Iran*

- Researched and developed machine-learning and deep-RL workflows for tax-evasion/fraud detection and anomaly analysis on operational financial data; validated model outputs and documented reproducible evaluations.

Network Expert / Systems Engineer **2018–2022**
Iranian National Tax Administration *Tehran, Iran*

- Administered and modernized network/server infrastructure supporting **1,000+ workstations across 30 cities**; worked with routing, switching, firewalls, Windows servers, virtualization, security, monitoring and incident response.
- Built automation and infrastructure utilities with Python, REST APIs, **Terraform** and Git; contributed Azure/AWS network designs, migration runbooks, capacity planning, and security hardening.

Software Expert – Senior Developer **2014–2018**
Iranian National Tax Administration *Rasht, Iran*

- Designed and developed **(back-end)** desktop and web applications for tax auditing, collection, and dispute-board workflows.

Cybersecurity Analyst & Programmer **2012–2014**
Gilan Science & Technology Park – Information Security Laboratory *Rasht, Iran*

- Developed end-point security software, analyzed malware/exploits, conducted penetration tests, and advised organizations on network/security protocols and defensive infrastructure.

Educational Activity

Undergraduate Instructor, Computer Engineering **2020–2023**
Azad University of Rasht *Iran*
Taught advanced programming, data structures and algorithms, automata theory, databases, software engineering, and artificial intelligence.

Bootcamp Educational & Technical Evaluator **2021–2023**
Maktab Sharif Coding School *Iran*
Designed periodic technical assessments and evaluated learning progress and bootcamp quality.

Selected Applied ML and Software Projects

— Back-end development for the Integrated Tax System auditing subsystem for **Iranian National Tax Administration**
— Identification of electricity-consumption patterns for unauthorized cryptocurrency mining for **Iran's Ministry of Energy**

Deep-RL autonomous-driving simulation and market prediction; LSTM-based stock forecasting; video captioning; brain-tumor detection; OpenCV-based vehicle lane detection; Hospital Information System.

Selected Honors and Professional Qualifications

- Ranked **10th among 13,000+** participants in the nationwide Computer Science M.Sc. entrance examination (2017); full tuition-waiver scholarship for Iranian national universities.
- Ranked **1st among 2,000+** candidates in the written exam and interview for Software Expert at the Iranian National Tax Administration (2014).
- Licensed Court Expert (IT), Department of Justice; Certified Software & Network Consultant, Iran ICT Guild Organization.
- Appreciation from the Director General of the Iranian National Tax Administration (2023); earlier national awards for software/technical innovations.

Languages

English: C1 (Advanced) | **French:** A2 (Elementary) | **Persian:** Native